

Technical Evaluation : Performance Benchmarking

1. Model Architecture & Specifications

This section defines the technical configuration of the vision-language models utilized in the OptiSight navigation pipeline.

Model Name	Backbone	Weight Class	Logic Tier (CoT)	Vision System	Primary Project Role
DeepSeek-VL-1.3B	DeepSeek	Feather	Basic	SigLIP	Fastest Reflex: Emergency Brake
Qwen2-VL-2B	Qwen-2	Feather	Basic	NaViT	Main Pilot: Real-time Nav
LLaVA-Phi-3-Mini	Phi-3	Light	Medium	ViT-L	Legacy Ref: Model Comparison
Phi-3.5 Vision	Phi-3.5	Light	High	ViT-L	Logic Expert: Rule Adherence
DeepSeek-Janus-7B	Janus	Mid-Heavy	High	SigLIP-L	Innovator: Architecture Testing
Qwen2-VL-7B	Qwen-2	Mid-Heavy	High	NaViT	Detail Eye: Distant Objects
LLaVA-Mistral	Mistral	Heavy	High	ViT-L	Industry Standard: Benchmark
LLaVA-Llama-3	Llama-3	Heavy	Superior	ViT-L	The Brain: Safety Analysis

2. Mathematical Performance Metrics (Theoretical vs. Practical)

The following metrics are used to calculate the feasibility of real-time deployment on Edge AI hardware.

2.1. Inference Latency Equations

The total time to decision (T_{total}) is calculated as:

$$T_{total} = T_{prep} + T_{inf} + T_{post}$$

Where:

- T_{prep} : Image pre-processing time (Kornia/Rust).
- T_{inf} : Model inference time (Time to First Token).
- T_{post} : Control command generation and motor latency.

2.2. Comparative Performance Data

Model Name	Weight Class	Latency (ms)	Throughput (TPS)	VRAM Usage	Power (Est.)	CoT Accuracy
DeepSeek-VL-1.3B	Feather	~25ms	52	1.2 GB	6W	62%
Qwen2-VL-2B	Feather	~40ms	38	1.9 GB	9W	68%
LLaVA-Phi-3-Mini	Light	~160ms	22	2.6 GB	18W	76%
Phi-3.5 Vision	Light	~190ms	20	3.1 GB	22W	84%
DeepSeek-Janus-7B	Mid-Heavy	~320ms	14	4.6 GB	40W	88%
Qwen2-VL-7B	Mid-Heavy	~360ms	13	4.9 GB	45W	90%
LLaVA-Mistral	Heavy	~480ms	10	5.3 GB	55W	91%
LLaVA-Llama-3	Heavy	~520ms	9	5.7 GB	62W	95%

3. Critical Analysis & Selection Logic

3.1. The Resolution Advantage (NaViT vs. ViT)

Traditional Vision Transformers (ViT) require images to be resized to a fixed square (336×336), causing spatial distortion. Qwen2-VL utilizes **NaViT (Native Resolution ViT)**, allowing for dynamic aspect ratios:

- **Impact:** Improved detection of thin/distant obstacles (e.g., fence wires, small rocks).

3.2. Reasoning Depth (CoT vs. Zero-Shot)

Large models (>7B) demonstrate a superior ability to handle "Exceptional Cases" through Chain-of-Thought (CoT).

- **Equation of Reliability:**

$$R = \frac{Intelligence(cot)}{Latency(T)}$$

- **Result:** While Llama-3 has higher latency, its reliability (R) for complex path-planning makes it indispensable for non-linear obstacle negotiation.